

Cloud Computing Platform

WORKSHOP OVERVIEW: MULTI-USER URL SHORTENER



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Content

Common aspects of Cloud Platform

Cloud Computing stack

“Taxonomy” of Cloud systems

Open Source Cloud computing platform

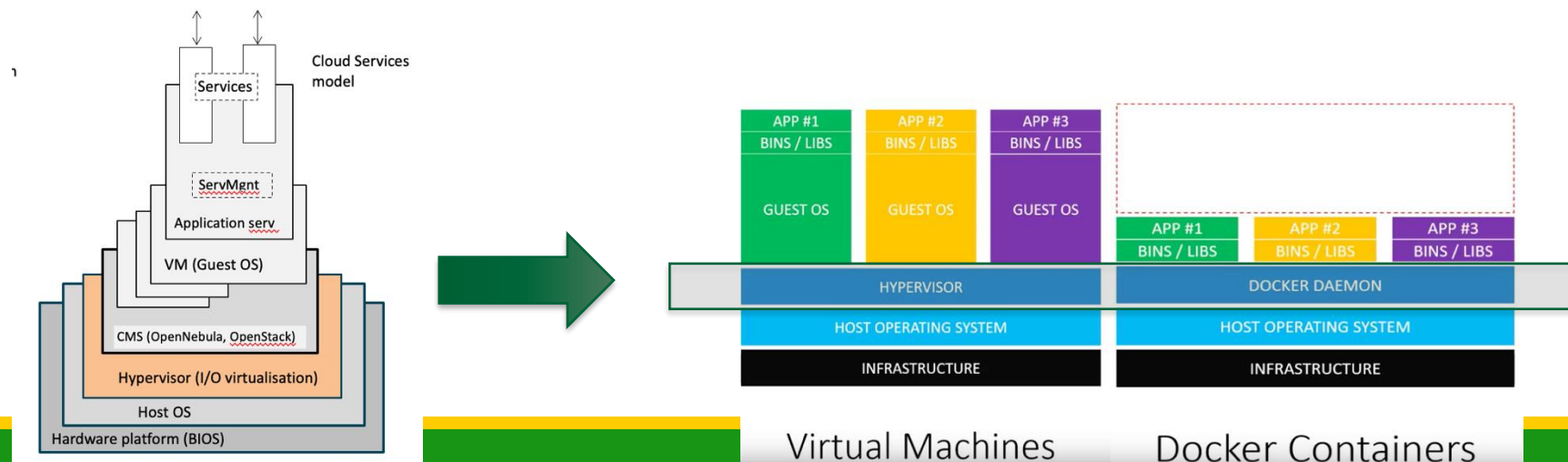


Design requirements for Cloud stack

Cloud stack: How to "Capitalize" on Virtualization

Transparency: How much the user sees of the underlying hardware.

Customization: what can the user change in the newly provided infrastructure ("Freedom vs. Automation" Trade-off)

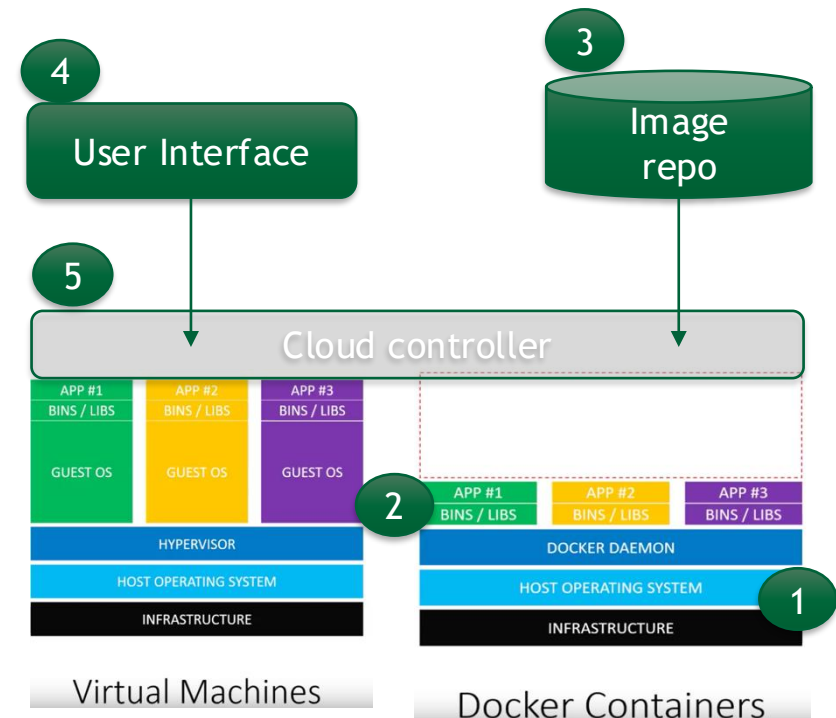




Cloud Computing stack

Cloud platform

1. Hardware, Operating systems and Network
2. Virtualization: VM & hypervisor / Containers
3. Image Catalog
4. User Interface





Interactions with Cloud platform: users/providers

Negotiation (User Interface)

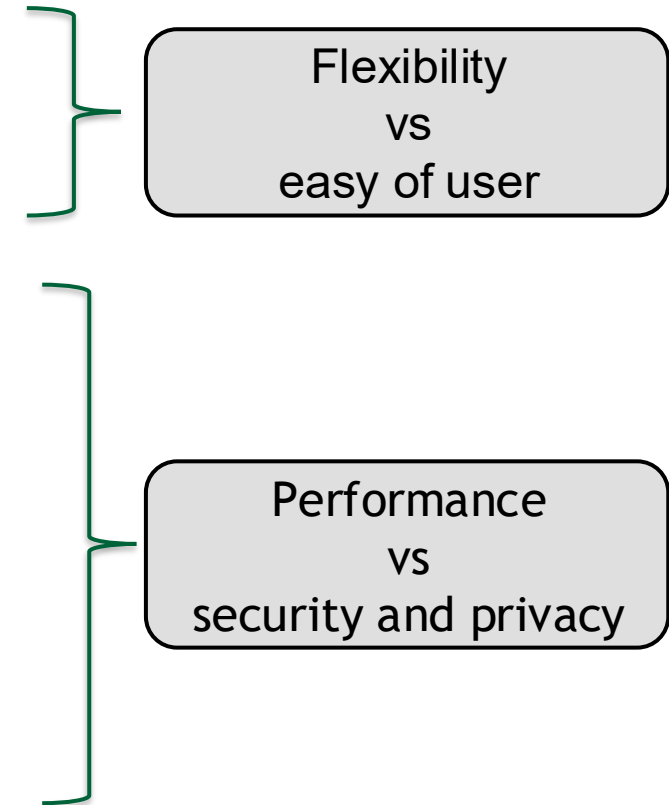
- **how users-developers interface** with the cloud

Decision (VM/containers placement)

- How **VM/Containers are scheduled** to meet users' requirements

Operation

- How to **enforce the decision** and **the communications** between cloud elements





Negotiation (User Interface)

In the cloud approach all functions as a services exposed through an **API**,

Implemented as : **Web Services/REST**

To enable Unified Programmability

Users code doesn't just run on the cloud; It can
“manage” its portion of the cloud

- how many resources Elastic growth,
- Type of OS, the runtime system ...



Decision (VM placement)

Users $\Leftrightarrow$ Providers

No best effort in Cloud:

- Cloud **providers** must fulfil the **SLA** signed with the users

It is optimization problem

Complex optimization:

- Resource Availability:
- Energy Efficiency:
- Affinity: speed vs safety

Availability, up time, elasticity, backup strategy, ...



Decision (VM placement)

From Virtualization to Orchestration

VMs/containers **need to communicate** using different elements of the Cloud stack:



Often Web services/**REST** are used to communicate with

- Storage
- compute elements

But also, **traditional traffic engineering protocols** are used to communicate/control network elements

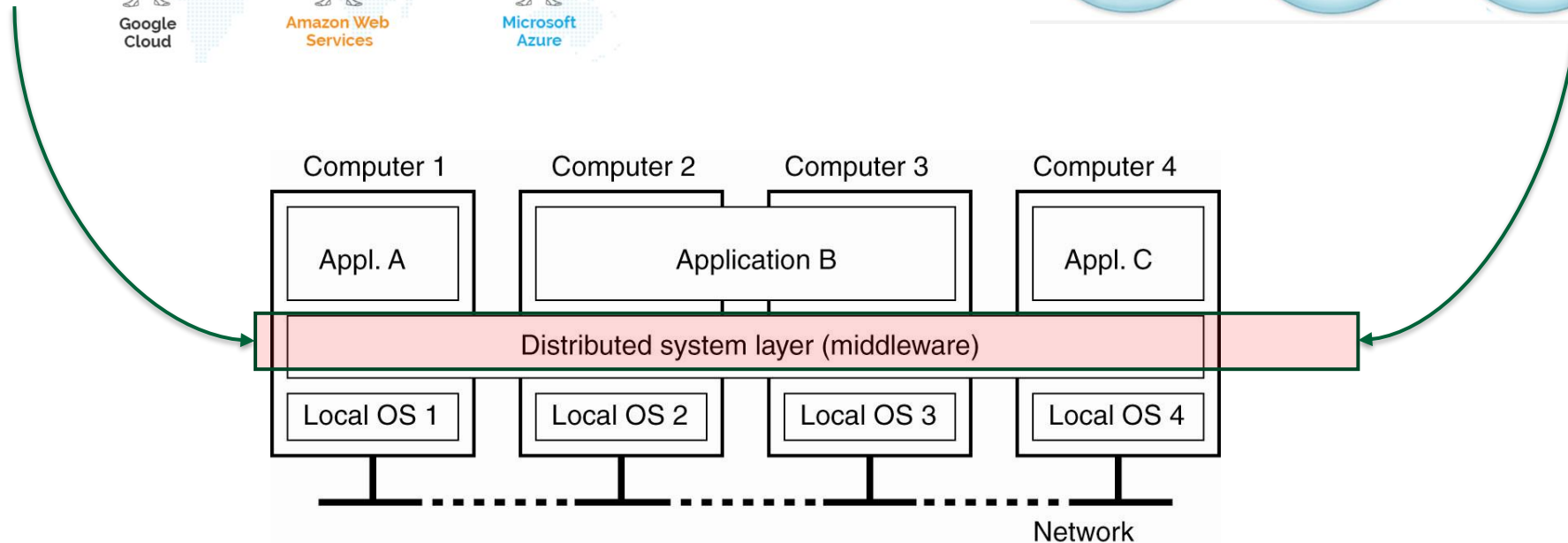
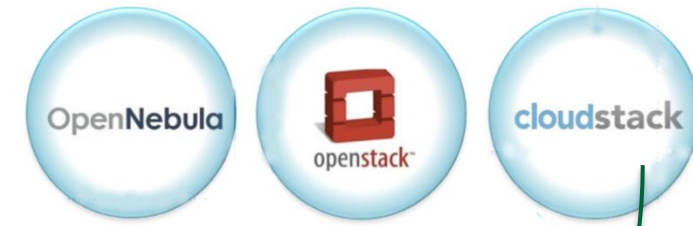


Cloud platforms

Closed source “public Cloud”



Open Source “private cloud”





Open Cloud platforms

Platform	Current Status	Primary Specialization / "Niche"	Why it Matters Today
OpenStack	Standard	Large-scale Enterprise & Telcos	" De-Facto " open-source standard for big data centers.
OpenNebula	Standard	Private Cloud & Edge Computing	Popular lightweight and easy to deploy.
Eucalyptus	Legacy	AWS API Compatibility	Pioneer : prove that REST APIs could make clouds interchangeable.
Nimbus / Chameleon	Specialized	Scientific Research	Specialist : "Science Clouds" where hardware needs to be "leased" for experiments.
Apache VCL	Specialized	Education / Lab Management	Perfect for university environments to reset student desktops.
CloudStack	Active	Service Providers	competitor to OpenStack focused on simplicity.



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Next Section



NIST-cloud Taxonomy

Level 1: **Role** → obligations / behaviors of associated actors in cloud computing.

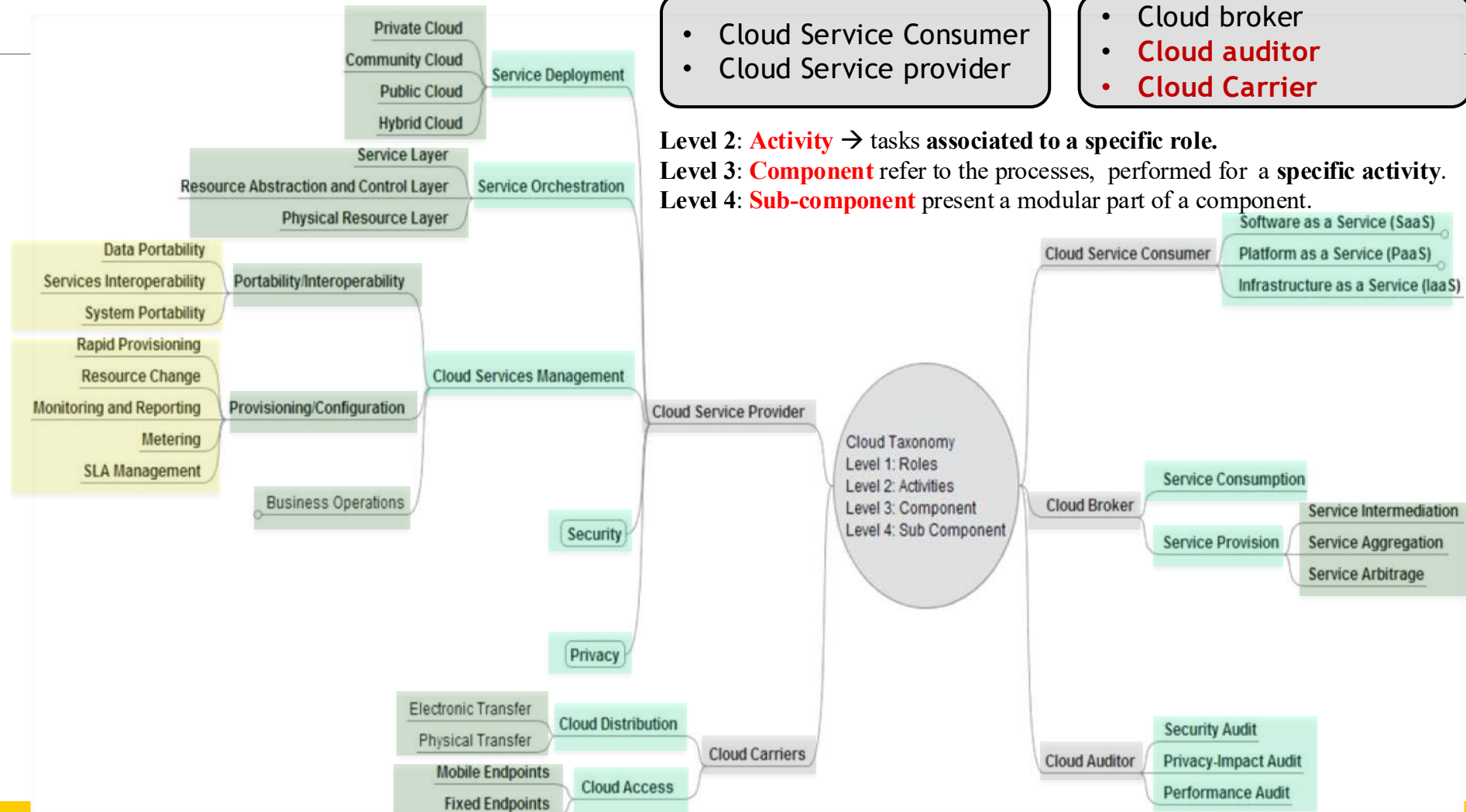
- Cloud Service Consumer
- Cloud Service provider

- Cloud broker
- **Cloud auditor**
- **Cloud Carrier**

Level 2: **Activity** → tasks associated to a specific role.

Level 3: **Component** refer to the processes, performed for a specific activity.

Level 4: **Sub-component** present a modular part of a component.





Content

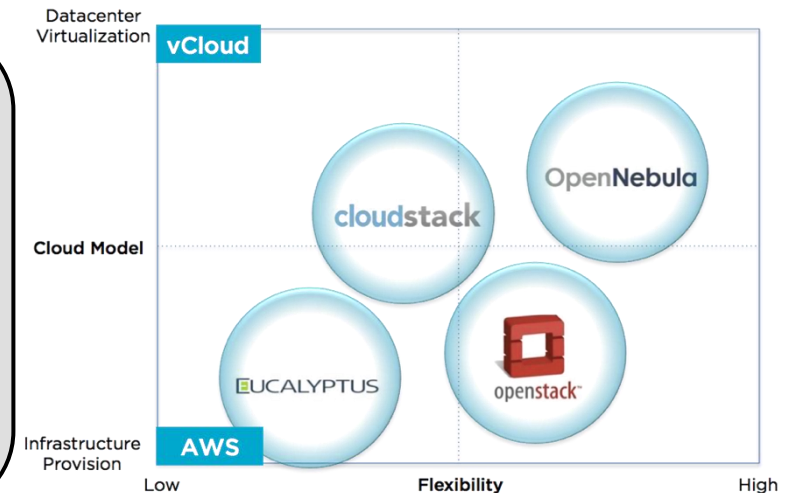
Common aspects of Cloud computing Platform

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Open-Source Cloud computing platform

- ☐ Cloud-burst (elastic provisioning)
- ☐ Multi-cloud / Federated-cloud
 - ☐ Federated ID management / security / resource management / data management
- ☐ Integration with Third-part software
- ☐ Edge Cloud continuum





OpenStack

The Most Widely Deployed Open Source Cloud Software in the World

Deployed by thousands. Proven production at scale. OpenStack is a set of software components that provide common services for cloud infrastructure.

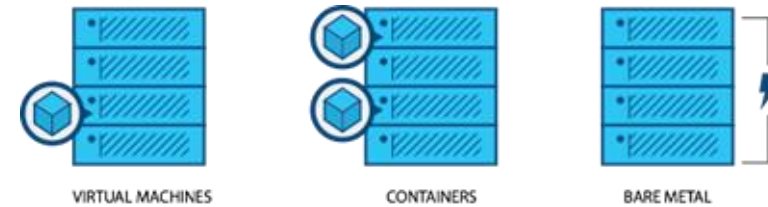
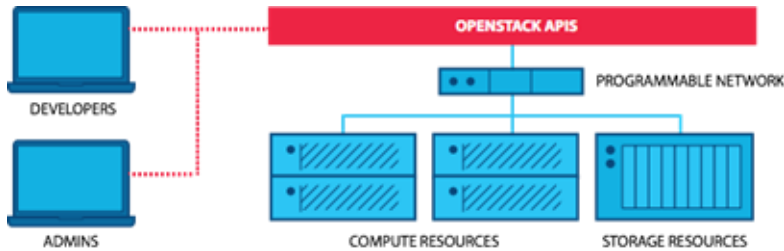
[BROWSE OPENSTACK COMPONENTS](#)

OpenStack is developed by the community. For the community. [Learn how to contribute →](#)



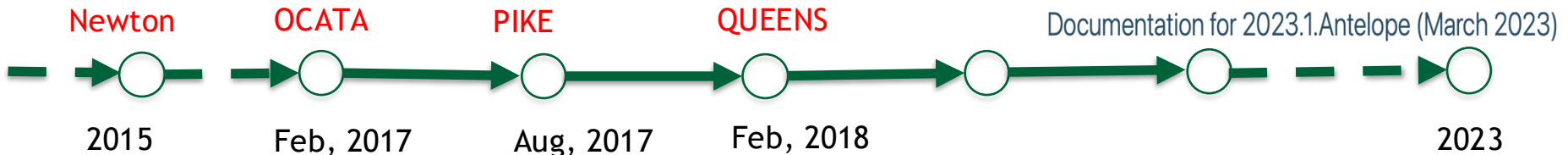


What is OpenStack



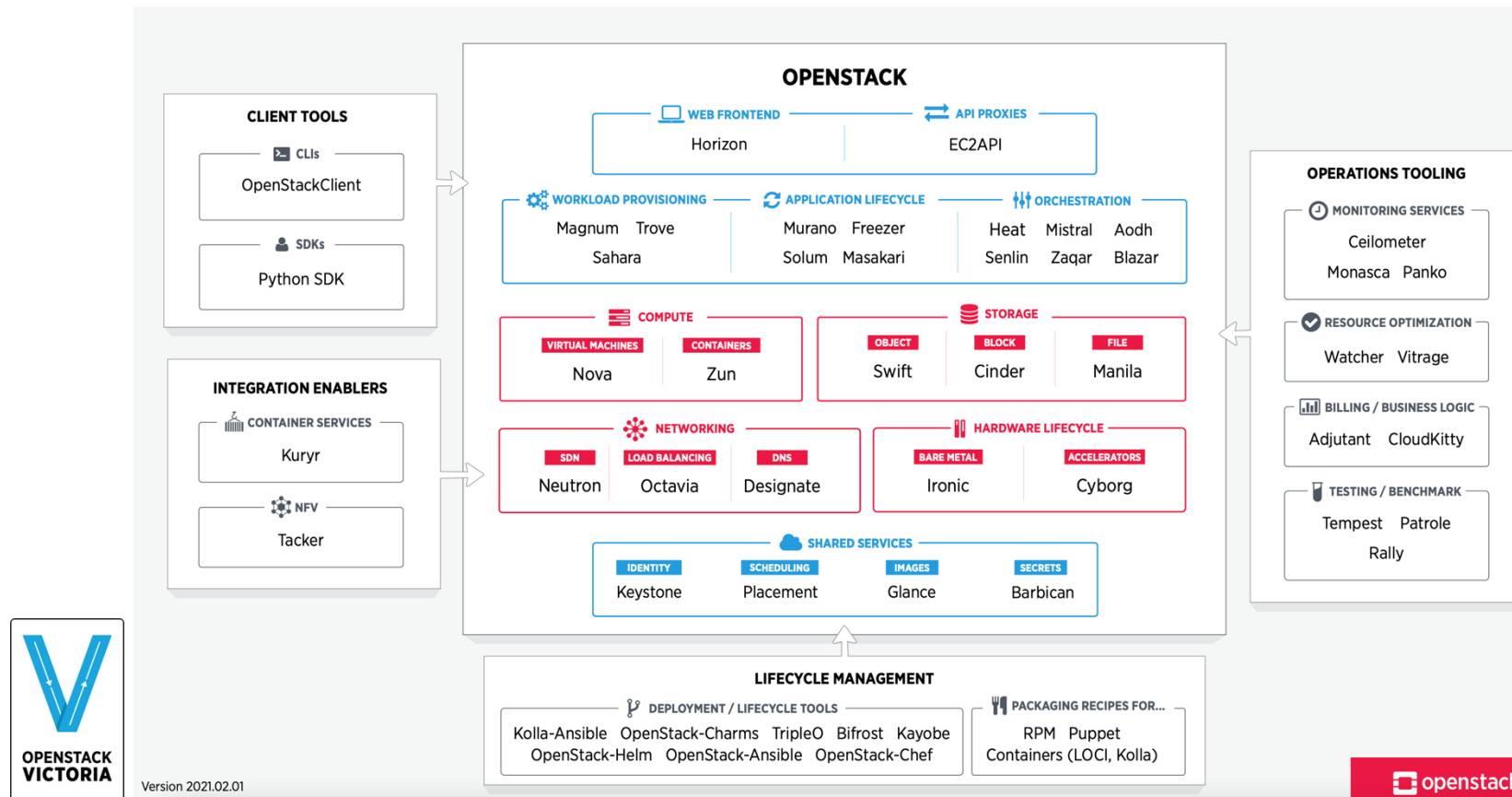
Programmable infrastructure that lays a common set of APIs on top of compute, networking and storage

One platform for virtual machines, containers and bare metal

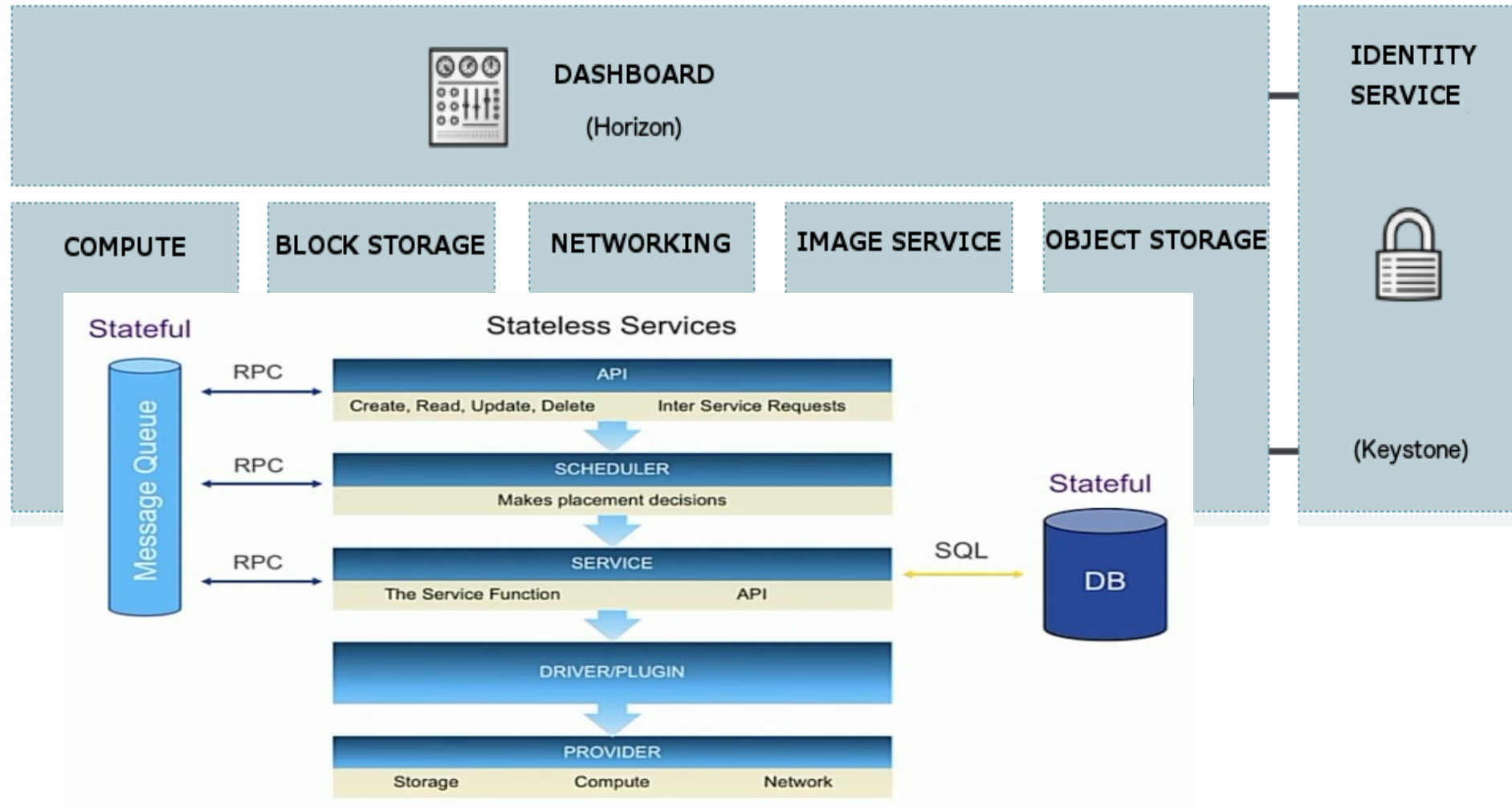




OpenStack project Map



OpenStack core services



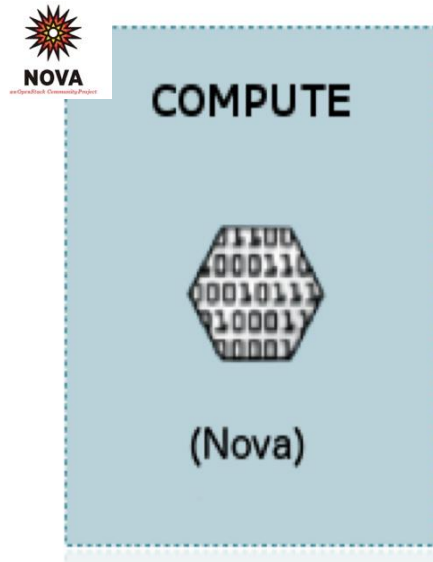
OpenStack core services: **Nova**



Core Function:

Massive Scale: Orchestrates self-service access to global compute resources.

Abstraction Layer: Manages VMs, Bare Metal, and Containers through a unified interface



Key Features:

- **Hypervisor Agnostic:** REST API compatible with KVM, VMware, Xen, and even Amazon EC2.
- **Message-Driven:** to allow horizontal scaling across nodes.
- **Automated Coordination:** Handles the full lifecycle (scheduling, creation, and migration) of virtual resources.

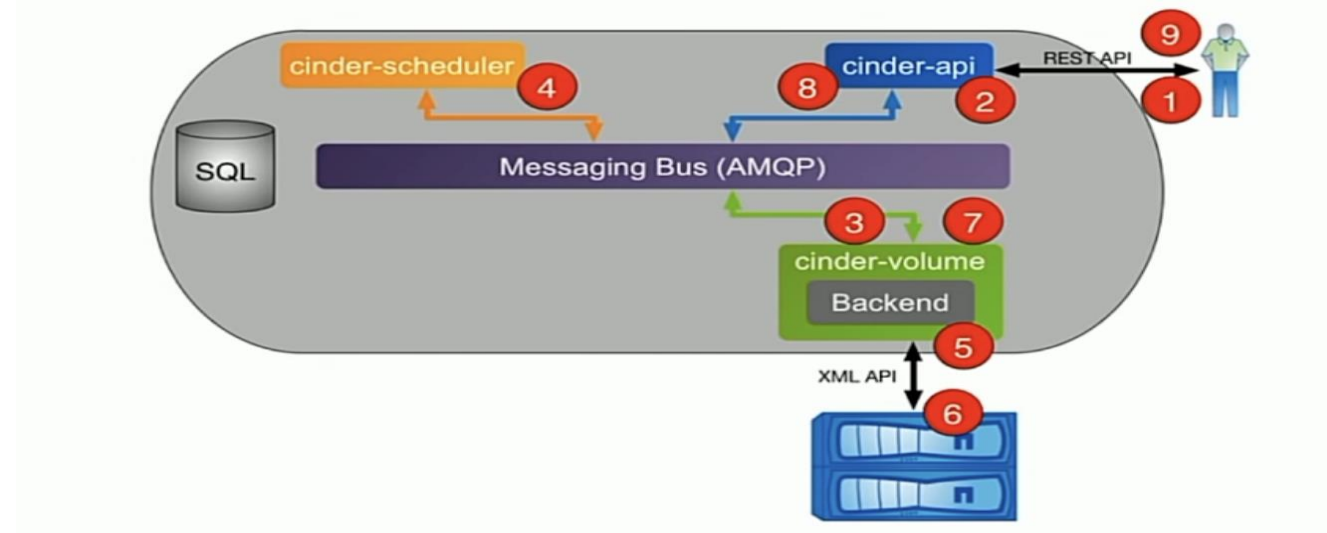
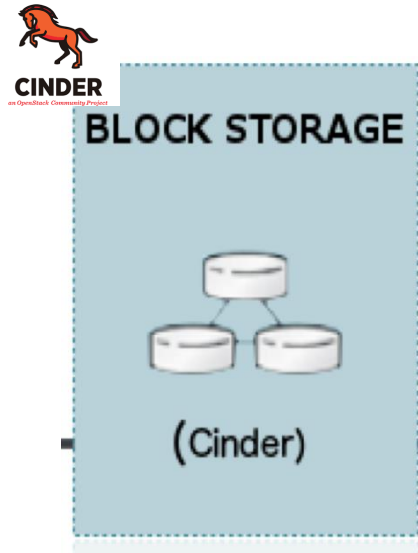


OpenStack core services: **Cinder**

Core Function:

Life Cycle Management: Manages persistent block storage—from creation to attachment (to Nova instances) and finally release.

Persistence: Unlike ephemeral storage, Cinder volumes survive VM reboots and deletions, making them essential for databases and user data.





OpenStack core services: **Neutron**

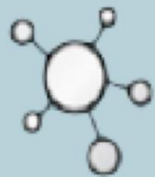
Core Function:

NaaS (Networking as a Service): Provides a programmable, API-driven interface to manage complex network topologies for virtual compute environments.

Complete Abstraction: Allows users to define their own Virtual Networks, Subnets, and Routers without touching physical hardware.



NETWORKING



(Neutron)

Key Features:

- **Decoupled Architecture:** Uses the same Topic-based Messaging to trigger network changes.
- **Software-Defined Networking (SDN):** Enables advanced features like Floating IPs (Elastic IPs), Security Groups (Virtual Firewalls), and Load Balancing.
- **Pluggable Plugins/Drivers:** Supports multiple technologies (Open vSwitch, Cisco, VMware NSX, Juniper) through a unified API.



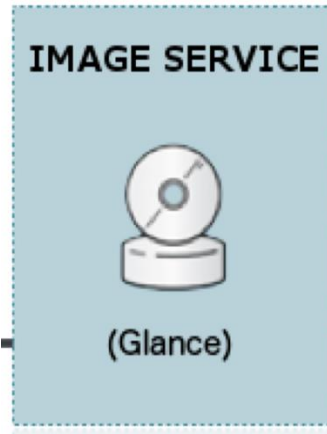
OpenStack core services: **Glance**

Core Function:

The Cloud Library: Acts as a central repository for discovering, registering, and retrieving Virtual Machine (VM) images.

Metadata Management: Stores detailed information about images (OS type, kernel versions, hardware requirements) to allow precise querying.

Key Features:



- **RESTful API:** Provides a standardized interface to query image metadata and stream the actual image data to Nova.
- **Storage Agnostic:** Does not store images itself; it uses backends like Swift, S3, or local filesystems.
- **Format Flexibility:** Supports a wide variety of disk formats, including:
 - **Standard:** RAW, QCOW2 (KVM/OpenStack default).
 - **Legacy/External:** VHD (Hyper-V), VMDK (VMware), VDI (VirtualBox), ISO, and AMI (Amazon).,

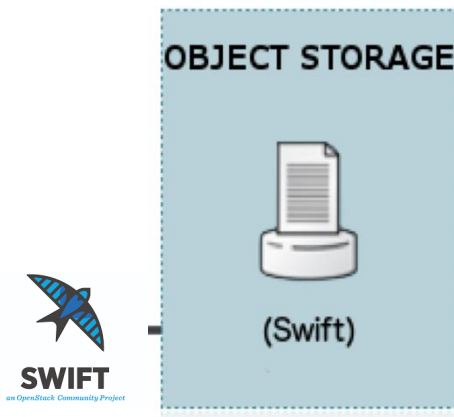


OpenStack core services: **Swift**

Core Function:

Object Blob Store: A distributed system for storing unstructured data (images, videos, backups) as "objects" rather than files in a hierarchy.

Massive Scale: Built to handle petabytes of data with high concurrency and durability across the entire cluster..



Features

- **Eventually Consistent:** Prioritizes availability;
- **High Availability:** No single point of failure; the system automatically detects and repairs corrupted data (self-healing).
- **S3 Compatibility:** Can be configured to use the Amazon S3 API, allowing seamless migration from public to private clouds.

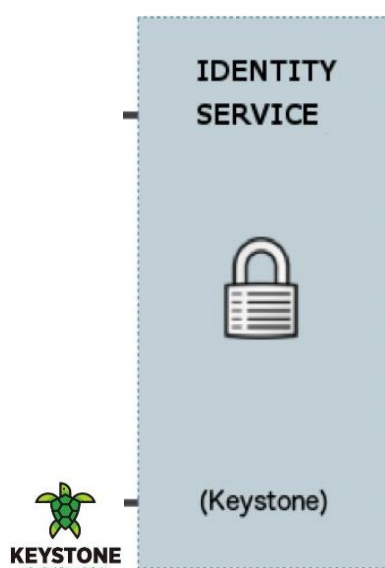


OpenStack core services:

keystone

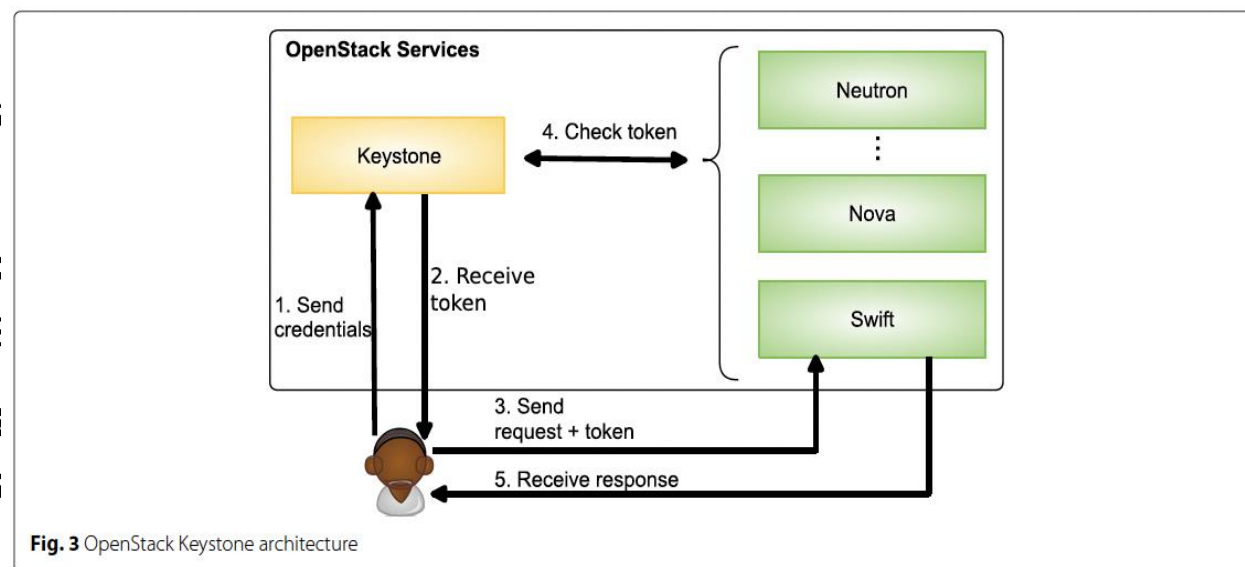
Keystone provides

- API client authentication,
- distributed multi-tenant authorization
- Keystone support: LDAP , Oauth, OpenID connect, SAML and SQL.

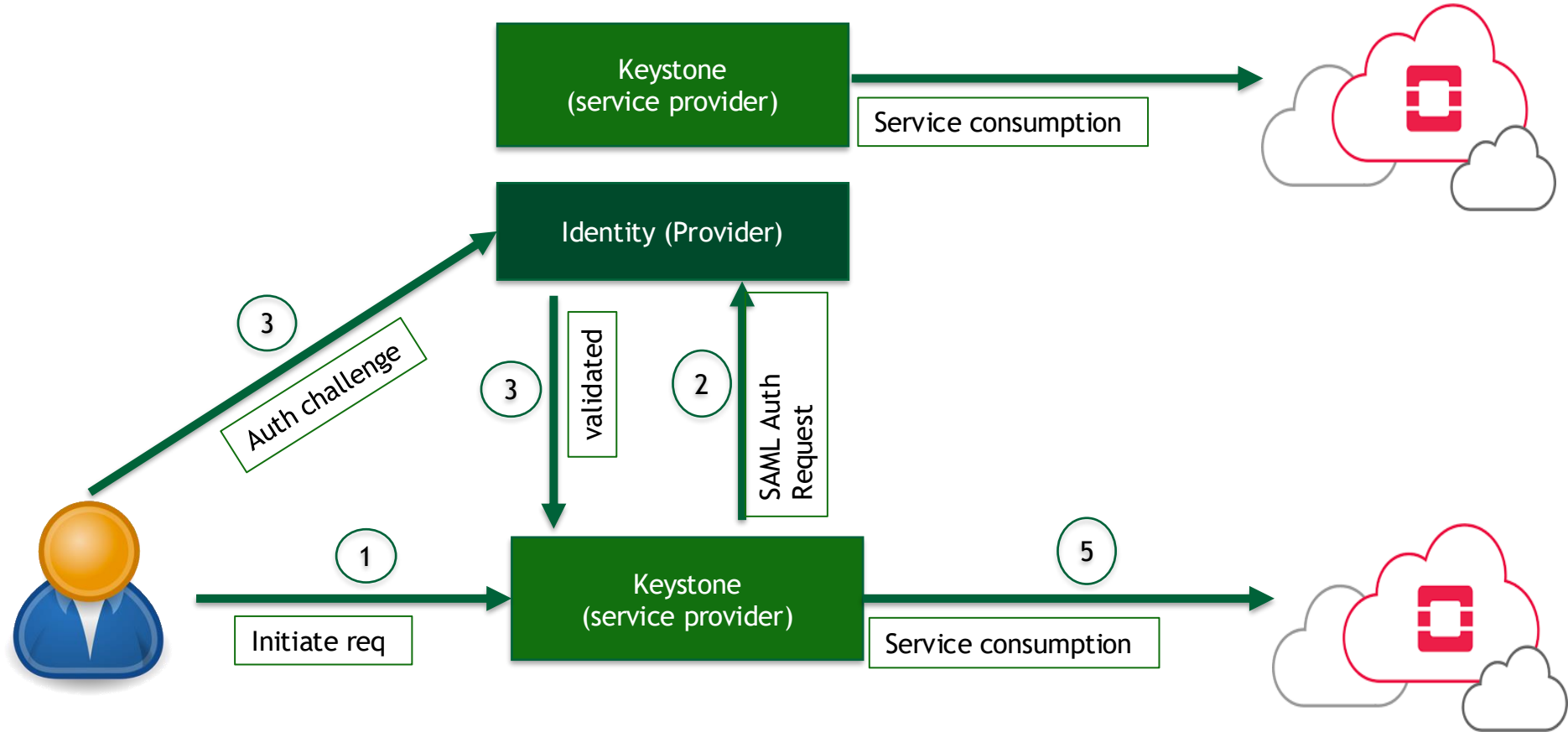


Simple processes
policy services

- **Token service**
other services
- **Service registration**
service exposure



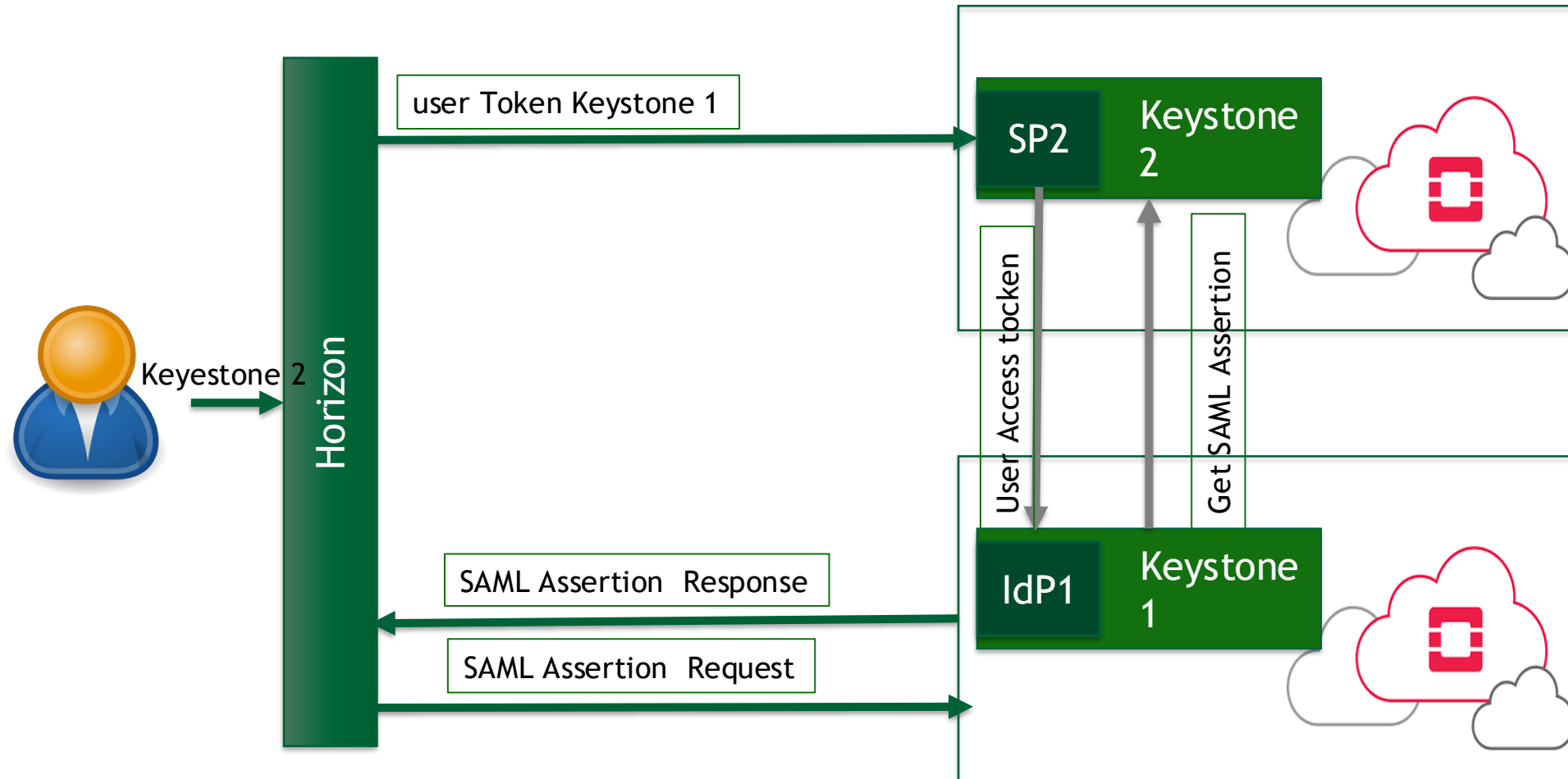
OpenStack identity management



Problem: What if I have multiple Keystone instances running, how would I create a federated identity management ?



Federated Identity Keystone-2-Keystone



Solution : Keystone federated identify management



Orchestration

Multi-cloud using OpenStack

```
heat_template_Version: 2012-0430
```

```
description:
```

```
resources:
```

```
  my_instance:
```

```
    type: OS::Nova:: Server
```

```
    properties:
```

```
      image: Standard_OpenSUSE_42.1_JeOS_Latest
```

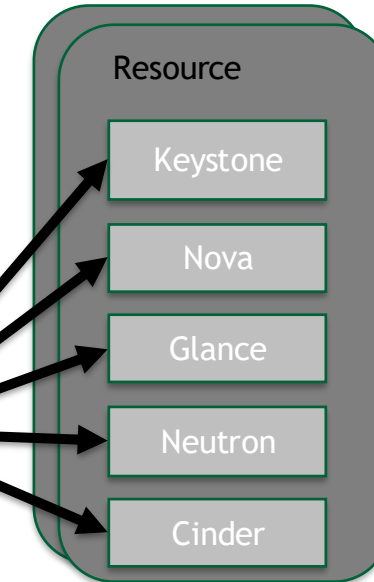
```
      flavor: C1.meduim
```

```
      key_name: my Key
```

```
      networks
```

```
        - networks: my network_Id
```

```
      user_data_format: RAW
```



Problem: how to connect two or more openstack cloud instances ?

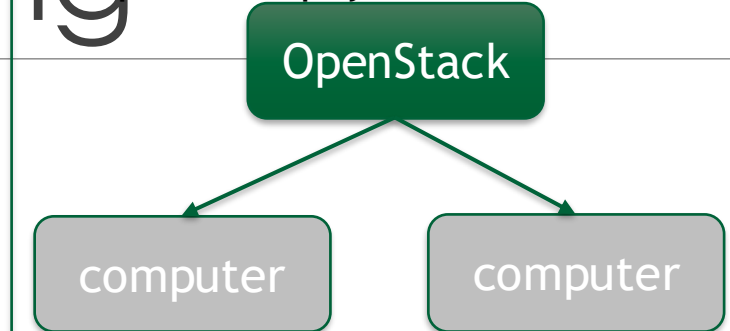


OpenStack cascading

OpenStack can manage many computers,

- why not treat OpenStack itself as one huge supercomputer?

One OpenStack deployment

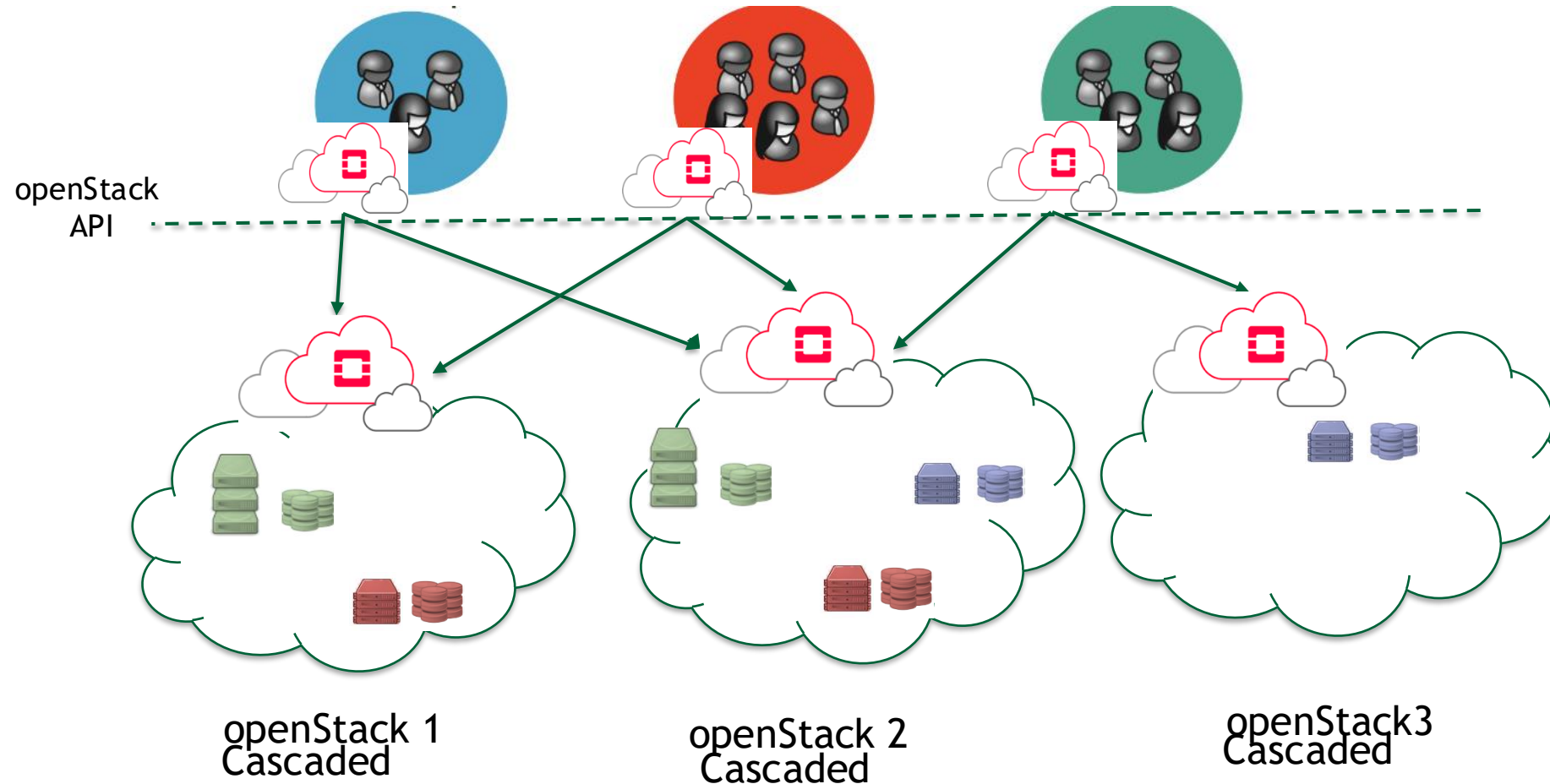


by the parent OpenStack

Problem: is there a different way of connecting multiple openStack instances?

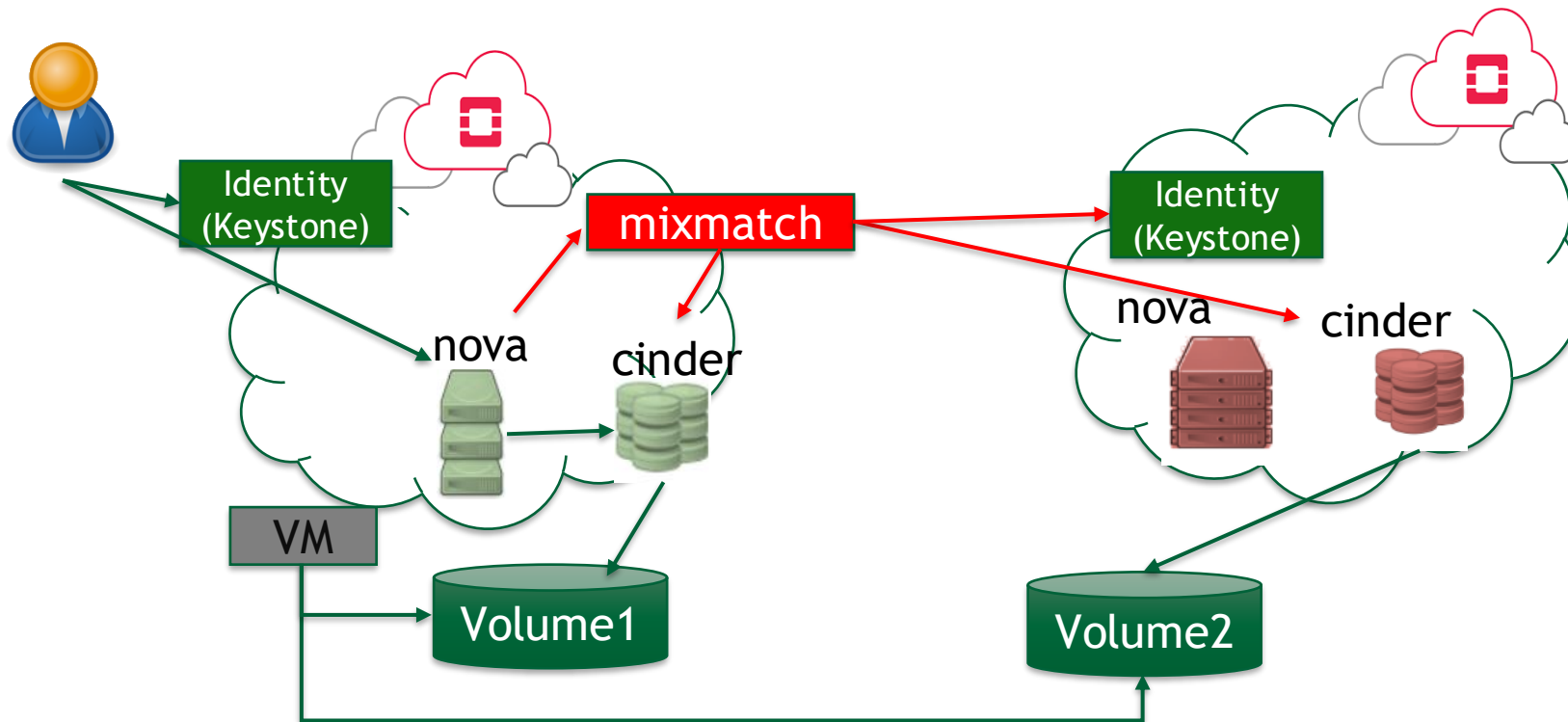


OpenStack cascading





Multi-cloud using OpenStack



Problem: is there a different way of connecting multiple openStack instances?



Bare Metal & TripleO

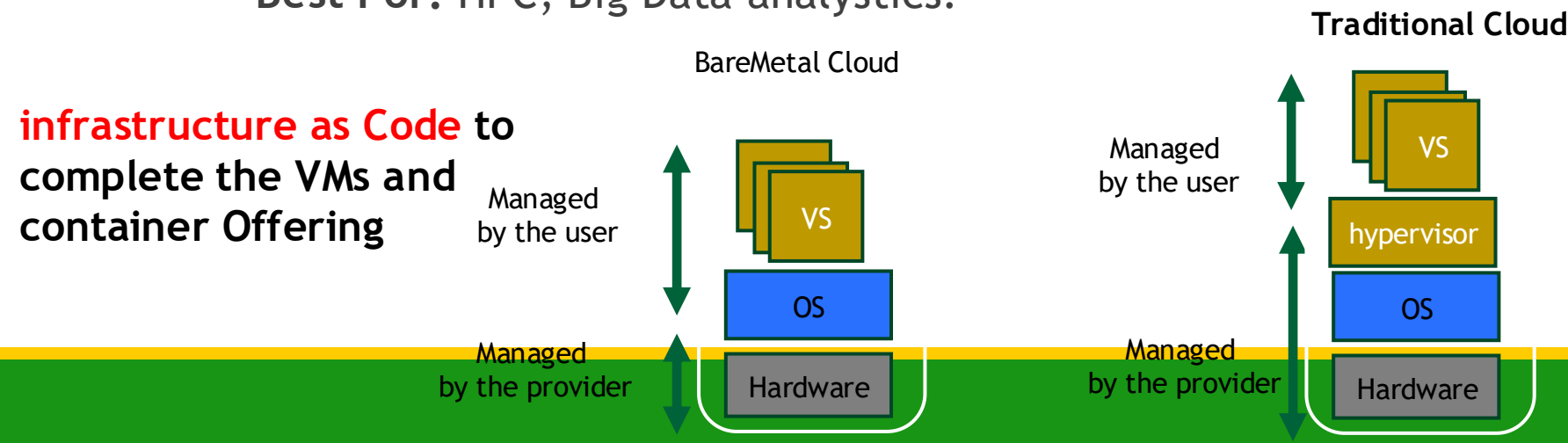
BARE METAL MODEL (OPENSTACK IRONIC)

- Provides physical servers to users without a hypervisor layer.
- The provider (OpenStack) manages the hardware lifecycle, but the user gets direct access to the physical CPU, RAM, and Disk.
- **Best For:** HPC, Big Data analytics.

TRIPLEO: THE "OPENSTACK-ON-OPENSTACK" INSTALLER:

• **Infrastructure as Code:** TripleO uses OpenStack to deploy OpenStack.

- **The Undercloud:** A small bare-metal management cloud used to "install" the production cloud.
- **The Overcloud:** The functional cloud that the end-user eventually sees.

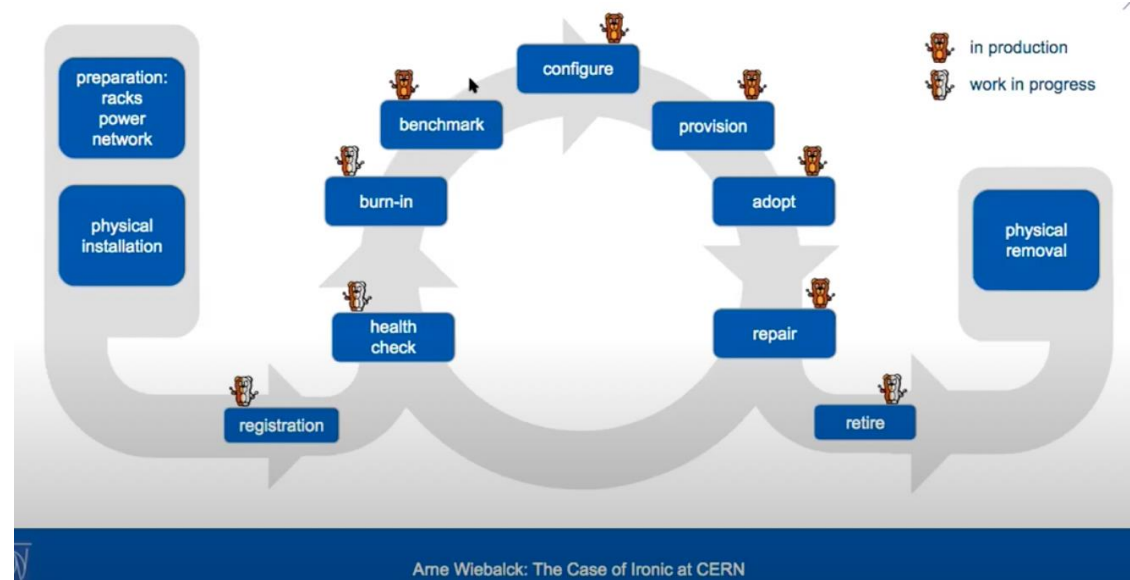




OpenStack ironic

open-source bare metal **provisioning framework** within the OpenStack cloud computing platform

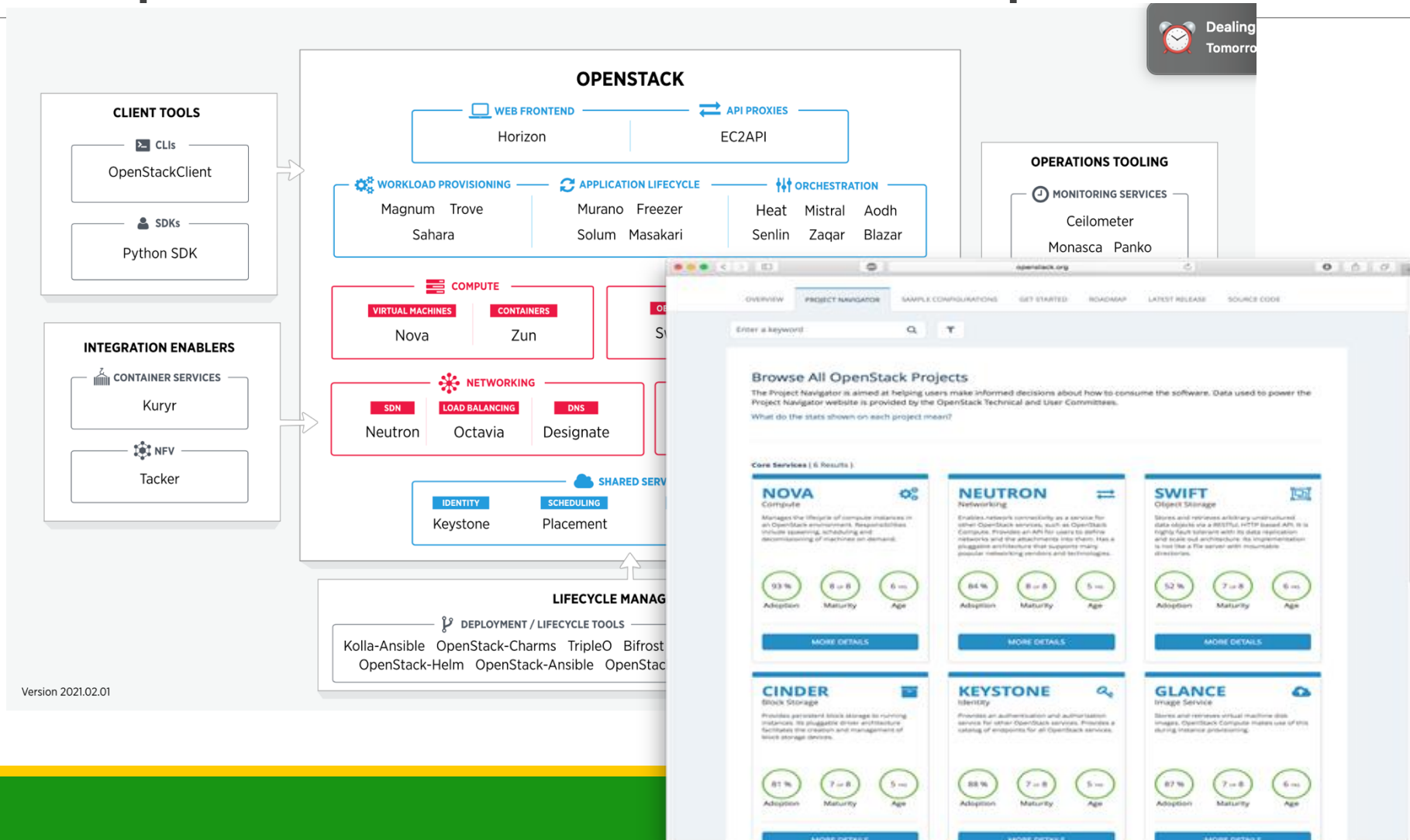
- Hardware discovery
- hardware introspection,
- power management,
- node management



through a RESTful API



The openstack Landscape





In Summary, the Cloud Operating System enables enterprises to

Control and **automate** pools of resources

Efficiently allocate resources

Empower admins & users via self-service portals

Empower developers to make apps cloud-aware via APIs

On behalf of the DIGITAfrica consortium



Thank you!